

PRAVIN KUMAR SINGH, Ph.D.



Nationality: Indian

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Google scholar: <https://scholar.google.com/citations?user=2c11XTIAAAAJ&hl=en>

Academic Qualification

Doctor of Philosophy (Ph.D.)

Department of Physics and Material Science,
Madan Mohan Malaviya University of Technology, Gorakhpur, India

May 2019

M.Tech. in Optoelectronics and Laser Technology with 82.35%

School of Electronics and Photonics
Pt. Ravishankar Shukla University Raipur, India

June 2015

M.Sc. in Electronics and Communication with 77.40%

School of Electronics
Devi Ahilya Vishwavidyalaya, Indore, India.

June 2012

B.Sc. with Physics, Electronics, Mathematics with 67.72%

D.D.U. Gorakhpur University, Gorakhpur, India

June 2010

Teaching Experience

Assistant Professor

Department of Physics
Manipal University Jaipur, Rajasthan, India

(1 Dec 2025—till date)

Assistant Professor

Department of Applied Science and Humanities
KIPM College of Engineering and Technology, Gorakhpur, India

(12 Oct 2021—31 March 2022)

Guest Faculty

Department of Physics and Material Science
M.M.M. University of Technology Gorakhpur, India

(01 Jan 2021—25 Sep 2021)

Guest Faculty

Department of Physics and Material Science
M.M.M. University of Technology Gorakhpur, India

(22 Jul 2019—31 May 2020)

Research cum Teaching Fellow

Department of Physics and Material Science
M.M.M. University of Technology Gorakhpur, India

(13 Jul 2015—13 May 2019)

Research Experience

Marie Skłodowska-Curie Individual Fellow (MSCA-IF)

Funded by European Commission
Institute of Advanced Materials, IAAM AB, Sweden
Research Interests: Smart crystal nucleation near interface dynamics in API

(1 July 2023— 30 November 2025)

Prof. Kobayashi Shunsuke Postdoctoral Fellow

KTH Royal Institute of Technology, Sweden
Research Interests: Development of self-powered wireless nanopiezotron for healthcare

(1 June 2023-30 June 2023)

Group Leader

VBRI Innovation Center
7-16 Under Ground Floor, Kalkaji Extension, New Delhi
Research Interests: Development of climate neutral flexible health and energy nanogenerator

(5 Apr 2022— 3 May 2023)

Ph.D. Research Scholar

Department of Physics and Materials Science, M.M.M. University of Technology Gorakhpur (U.P.), India.
Title of Ph.D. Thesis: Synthesis and characterization of selenium based chalcogenide glasses for optoelectronic applications

(13 July 2015—13 May 2019)

M.Tech. Research Project

(July 2014—Jun 2015)

Laser Plasma Division, Raja Ramanna Centre for Advanced Technology, Indore, India.
Title of project: Equation of state of aluminum using high power Nd: Glass laser system

M.Sc. Research Project

(Dec 2011—Jun 2012)

Devi Ahilya Vishwavidyalaya, Indore, India.
Title of project: Arduino Based Photovore Robot

Research Areas

- Chalcogenide based materials, for phase change memory.
- Photovoltaic cell for energy harvesting.
- Self-powered high performance piezo/tribo/pyro/electric nanogenerators (PENGs/TENGs/PyENG).
- SPR sensor/actuator.
- Advanced nanomaterials (2D) and devices for energy harvesting/sensing/storage applications.

Experimental Skills and Proficiencies

Instruments used and operated extensively:

- **Synthesis:** Nanofabrication, Melt Quenching
- **Thin-film deposition technique:** Spin coating, Thermal Evaporation, Chemical Bath Deposition
- **Piezoelectric/Triboelectric Energy Harvesting Characterization:** Digital Oscilloscope, Electrometer, source meter, d_{33} piezometer for evaluation of piezoelectric coefficients (d_{33} , d_{31} and d_{15}) and hysteresis loop tester for P-E, C-V, Strain measurement.
- **Electrical Measurement:** Impedance Spectroscopy, LCR meter, Autolab Galvanostate/Potentiostate
- **Structural, Thermal and Optical Characterization:** XRD, FTIR, Raman, UV-VIS, FESEM, AFM, and DSC.
- **Computational Simulation:** SCAPS, COMSOL, MATLAB, TANNER TOOL, ARDUINO.
- **Pellet machine:** Hydraulic machine

Academic Awards & Achievements

- **Humboldt Research Fellowship for Experienced Researchers**, Alexander von Humboldt Foundation, Germany
- **IAAM Scientist Award**, 63rd Assembly of Advanced Materials Congress, Sweden, on 25-27 March 2024
- **Marie Skłodowska-Curie Individual Fellowship**, Institute of Advanced Materials, IAAM AB, Sweden, on 1 July 2023, Funded by European Commission
- **Prof. Kobayashi Shunsuke Postdoctoral Fellowship**, KTH Royal Institute of Technology, Sweden, 1 June 2023, Funded by International Association of Advanced Materials, Sweden
- **Research cum Teaching Fellowship**, MMM University of Technology, Gorakhpur, India (July 2015-May 2019).
- **Best poster presentation award** in TEQIP-III sponsored Research Scholar Day (RSD-2018) organized by Department of Applied Science, M.M.M. University of Technology, Gorakhpur (17/12/2018).
- **Best paper award** in 5th IEEE Uttar Pradesh Section International Conference on Electrical, Electronics and Computer Engineering (UPCON), Jointly organized by M.M.M. University of Technology, Gorakhpur (U.P.) and University of Ryukyus, Okinawa, Japan (2/11/2018-4/11/2018).
- **Best poster presentation award** in TEQIP-III sponsored Research Scholar Day (RSD-2017) organized by Department of Applied Science, M.M.M. University of Technology, Gorakhpur (12/12/2017).
- **Best poster presentation award** in Malaviya Research Conclave (MRC) 2018 held at M.M.M. University of Technology, Gorakhpur, India (6/7/2018-8/7/2018).
- **First Division with Distinction** in Master of Technology.
- **First Division with Distinction** in Master of Science and Master of Technology.

Language Skills

English (Fluent), Hindi (Native language)

Major Courses Undertaken

Coursework completed during Ph.D.

1. Research Methodology 2. Advanced Mathematics 3. Advanced Microprocessor and its Applications 4. Advanced digital communication

List of Projects Completed /Ongoing

1. Major Research Project entitled “Energy Harvesting in Space Station using Nitrile Rubber based Triboelectric Nanogenerators” submitted to Vikram Sarabhai Space Centre, Thiruvananthapuram, Indian Space Research Organization, India
(Role: Principal Investigator, Application Number: RES-VSSC-2025-011, Status: **Submitted**, Project Cost: ₹ 57,05,000).
2. Major Research Project entitled “Smart Interface Engineering of Perovskite-Lignin based TENG for Sustainable Rural Electrification in India” submitted to Indian Council of Social Science Research, India
(Role: Principal Investigator, Application Number: 22773518, Status: **Submitted**, Project Cost: ₹ 25,28,000).
3. Major Research Project entitled “Smart Crystal Nucleation Near Interface Dynamics In Active Pharmaceutical Ingredients” funded by the European Commission.
(Role: MSCA IF, Grant Number: 101109228, Status: **Completed**, Duration: 1 July 2023-30 June 2025, Project Cost: 219,669.68 Euro).
4. Major Research Project entitled “Biohybrid Piezoelectric Nanogenerator for IoT Devices (BPN-IoT)” funded by VBRI Group Sweden.
(Role: Principal Investigator, Grant Number: 22773518, Status: **Completed**, Duration: 16 May 2022-15 May 2024, Project Cost: ₹ 25,00,000).

5. Major Research Project entitled “Quantum-Driven Optimization of High-Efficiency Perovskite Solar Cells for Sustainable Energy Harvesting (Q-HEPSC)” funded by VBRI Group Sweden.
(Role: Principal Investigator, Grant Number: 22773518, Status: **Completed**, Duration: 9 Apr 2022-08 Apr 2024, Project Cost: ₹ 27,00000).

List of Publications

Publications

1. Ashima Thakur, Prabal Pratap Singh, **Pravin Kumar Singh***, Ajit Sharma, Enhanced visible-light driven photocatalytic degradation of 2-chlorophenol using microwave-synthesized CoO/ZnO nanocomposites in aqueous media, *Pure and Applied Chemistry* (Accepted) 2026 (Q2, IF=2, SCIE)
2. Sujata Singh, Gun Anit Kaur, Mamta Shandilya, Ajit Sharma, **Pravin Kumar Singh***, Bandgap engineering and high-throughput defect dynamics in two-terminal CsGeI₃-Si monolithic tandem solar cells, *Scientific Reports* (Accepted) 2026 (Q1, IF=4.3, SCIE)
3. Kaushiki Sinha, Krishna Kumar Bhargav, Sadanand Maurya, **Pravin Kumar Singh***, Advanced Optimization of Lead-Free FASnI₃ Photovoltaics with Spiro-OMeTAD/ZnO Structure for Achieving 33.40% Power Conversion Efficiency, *Next Energy*, 12 (2026) 100713.
<https://doi.org/10.1016/j.nxener.2026.100713> (Q1, IF=6.4, SCIE)
4. **Pravin Kumar Singh***, Ajit Sharma, DK Dwivedi, The silent revolution: Quantitative perspectives on acoustic energy harvesting for self-powered micro- and nanosystems, *Solid State Communications* 415 (2026) 116464. <https://doi.org/10.1016/j.ssc.2026.116464> (Q2, IF=2.4, SCIE)
5. Divakar Singh, Surbhi Agarwal, D. K. Dwivedi, Pooja Lohia, Rajesh Kumar Yadav, **Pravin Kumar Singh**, and Upendra Kulshrestha, Exploring optical parameters of GeTe_(2-x)(SeSb)_x (x = 0, 0.2, 0.4, 0.6) Chalcogenide alloys for optical data storage, *Russ. J. Inorg. Chem.* 70 (2025) 2107-2120.
<https://doi.org/10.1134/S0036023625602247> (Q3, IF=1.5, SCIE)
6. Akhileshwar Mishra, D. K. Dwivedi*, & **Pravin Kumar Singh***, The physics and promise of phase change memory in non-von Neumann computing and machine learning. *Journal of Materials Chemistry C* 13(41) (2025) 20825-20861.
<https://doi.org/10.1039/D5TC02344K> (Q1, IF=5.1, SCIE)
7. Oishika Chatterjee, Gun Anit Kaur#, Nutan Shukla#, Sapna Balayan#, **Pravin Kumar Singh**#, Subhrangsu Chatterjee, and Ashutosh Tiwari, Multifaceted arsenal in SELEX nanomedicine." *Advances in Colloid and Interface Science* 342 (2025) 103540.
<https://doi.org/10.1016/j.cis.2025.103540> (Q1, IF=16, SCIE)
8. **Pravin Kumar Singh** and Upendra Kulshrestha, Unlocking the solar potential: elevating open circuit voltage in kesterite CZTS thin film solar cells through cutting-edge SCAPS modelling, *Eng. Res. Express* 7 (2025) 015301.
<https://doi.org/10.1088/2631-8695/ada33f> (Q2, IF=1.5, ESCI)
9. Akash Anand Verma, D.K. Dwivedi, Pooja Lohia, **Pravin Kumar Singh**, Rajesh Kumar Yadav, Manish Kumar, Surbhi Agarwal, Upendra Kulshrestha, Achieving 31.16% Efficiency in Perovskite Solar Cells via Synergistic Dion-Jacobson 2D-3D Layer Design, *Journal of Alloys and Compounds* 1010, 2025, 177882.
<https://doi.org/10.1016/j.jallcom.2024.177882> (Q1, IF=5.8, SCIE)

10. Anshuman Mishra, Marisa A. Frechero, Arnaud Caron, **Pravin Kumar Singh**, and Ashutosh Tiwari, Recent progress and future directions in nanoglass materials: A deep insight into synthesis, characterization, and application, *Nanotechnology and Precision Engineering* 8(1) 2025 015002-1-31.
<https://doi.org/10.1063/10.0028755> (Q1, IF=3.5, ESCI)
11. **Pravin Kumar Singh**, Soumyoraj Mallick, Gun Anit Kaur, Sapna Balayan, and Ashutosh Tiwari. John B. Goodenough's Pioneering Contributions towards Advancements in Photo-Rechargeable Lithium Batteries, *Nano Energy* (2024) 109792.
<https://doi.org/10.1016/j.nanoen.2024.109792> (Q1, IF=16.8, SCIE)
12. Tanmay Tiwari, Gun Anit Kaur, **Pravin Kumar Singh**, Sapna Balayan, Anshuman Mishra, and Ashutosh Tiwari, Emerging bio-capture strategies for greenhouse gas reduction: Navigating challenges towards carbon neutrality, *Science of The Total Environment* (2024) 172433.
<https://doi.org/10.1016/j.scitotenv.2024.172433> (Q1, IF=8.2, SCIE)
13. Neetesh Soni, **Pravin Kumar Singh**, Soumyoraj Mallick, Yash Pandey, Sreyansh Tiwari, Anshuman Mishra, and Ashutosh Tiwari, Advancing sustainable energy: Exploring new frontiers and opportunities in the green transition, *Advanced Sustainable Systems* (2024) 2400160.
<https://doi.org/10.1002/adsu.202400160> (Q1, IF=6.5, SCIE)
14. Anshuman Mishra, **Pravin Kumar Singh**, Nidhi Chauhan, Souradeep Roy, Ayushi Tiwari, Shaivya Gupta, Aanshi Tiwari et al. Emergence of integrated biosensing-enabled digital healthcare devices. *Sensors & Diagnostics* 3(5), (2024) 718-744.
DOI: [10.1039/D4SD00017I](https://doi.org/10.1039/D4SD00017I) (Q2, IF=3.5, ESCI)
15. Anshuman Mishra[#], Temitayo Omoyeni[#], **Pravin Kumar Singh**[#], S. Anandakumar, and Ashutosh Tiwari, Trends in sustainable chitosan-based hydrogel technology for circular biomedical engineering: A review, *International Journal of Biological Macromolecules* (2024): 133823.
<https://doi.org/10.1016/j.ijbiomac.2024.133823> (Q1, IF=7.7, SCIE)
16. D. Singh, S. Agarwal, D. K. Dwivedi, P. Lohia, R. Raj, & **Pravin Kumar Singh**, Optical and Glass Stability Parameters of Ge-Te-Se-Sb Chalcogenide Glasses for Optical Data Storage. *Russian Journal of Inorganic Chemistry* (2024) 1-16.
<https://doi.org/10.1134/S0036023623603501> (Q3, IF=1.8, SCIE)
17. Sadanand, **Pravin Kumar Singh**, Rajesh Tripathi, R.K. Yadav, D.K. Dwivedi, Hydrothermal synthesis of kesterite CZTSSe nanoparticles for highly efficient thin film, *Materials Chemistry and Physics* 320 (2024) 129434.
<https://doi.org/10.1016/j.matchemphys.2024.129434> (Q1, IF=4.3, SCIE)
18. Warren Rosario, **Pravin Kumar Singh**, Ashutosh Tiwari, Utkarsh Jain, D. K. Avasthi, and Nidhi Chauhan, Nanomaterial-based VOC sensing applications and a deep dive into their developmental trends, *Journal of Materials Chemistry A* 12 (2024) 9979.
<https://doi.org/10.1039/D3TA07171E> (Q1, IF=10.7, SCIE)
19. Tim Mashangva, Anshika Goel, Upasana Bagri, Sangeeta Prasher, Ajit Sharma, Mukesh Kumar, and **Pravin Kumar Singh**^{*}, Frontiers in MXene research: Pioneering synthesis, unveiled properties, and emerging applications in VOC detection, *Applied Materials Today* 38 (2024) 102163.
<https://doi.org/10.1016/j.apmt.2024.102163> (Q1, IF=7.2, SCIE)
20. Surbhi Agarwal, D.K. Dwivedi, Pooja Lohia, **Pravin Kumar Singh**, Calorimetric studies of yttrium doped non-conventional phase-change materials for improved performance, *Ceramic International*, 50(3) (2024) 5483-5497.
<https://doi.org/10.1016/j.ceramint.2023.11.301> (Q1, IF=5.1, SCIE)

21. Kinjjal Pandey, Pratap Kumar Dakua, Deepak Kumar Panda, **Pravin Kumar Singh**, and B. K. Pandey, Investigation of suitable buffer layer for PbS-TBAI/PbS-EDT colloidal quantum dot solar cell using SCAPS-1D, *Optik* 298 (2024) 171593.
<https://doi.org/10.1016/j.ijleo.2023.171593> (Q2, IF=3.1, SCIE)
22. **Pravin Kumar Singh**, Gun Anit Kaur, Mamta Shandilya, Poonam Rana, Radheshyam Rai, Yogendra Kumar Mishra, Mikael Syväjärvi, Ashutosh Tiwari, Trends in Piezoelectric Nanomaterials towards Green Energy Scavenging Nanodevices, *Materials Today Sustainability* 24 (2023) 100583.
<https://doi.org/10.1016/j.mtsust.2023.100583> (Q1, IF=7.1, SCIE)
23. Sharanya Kandula, **Pravin Kumar Singh**, Gun Anit Kaur & Ashutosh Tiwari, Trends in smart drug delivery systems for targeting cancer cells, *Materials Science and Engineering: B*, 297 (2023) 116816.
<https://doi.org/10.1016/j.mseb.2023.116816> (Q1, IF=3.9, SCIE)
24. Xuli Cheng, **Pravin Kumar Singh**, Anshuman Mishra, Ashutosh Tiwari, Wei Ren, Wurtzite nanostructured piezoelectric devices, *Materials Today Sustainability*, 23 (2023) 100474.
<https://doi.org/10.1016/j.mtsust.2023.100474> (Q1, IF=7.1, SCIE)
25. Shivam Singh, Sachin Singh, **Pravin Kumar Singh**, Rajesh Kumar Yadav, Pooja Lohia, D. K. Dwivedi, Theoretical study of malaria detection in blood samples using bimetal layer and zinc telluride nanomaterial-based surface plasmon resonance biosensor, *Plasmonics* 18 (2023) 1-12.
<https://doi.org/10.1007/s11468-023-01913-x> (Q2, IF=3.3, SCIE)
26. Rajnish Raj, Pooja Lohia, D. K. Dwivedi, **Pravin Kumar Singh**, Arpit Verma, B. C. Yadav, High performance photodetection sensors based on $(\text{S}_2\text{Ge})_{100-x}(\text{S}_3\text{Sb}_2)_x$ ($x = 15, 30, 45, 60$) system for optoelectronics applications, *Journal of Materials Science: Materials in Electronics* 34 (2023) 948.
<https://doi.org/10.1007/s10854-023-10359-0> (Q2, IF=2.8, SCIE)
27. Sachin Singh, Sushant Pandey, Sapna Yadav, Rajesh Kumar Yadav, **Pravin Kumar Singh**, Pooja Lohia, D. K. Dwivedi, Numerical Study Among Au, Al, and Ag Metal based Surface Plasmon Resonance Sensor, *Journal of Optics*, 53 (2024) 304-314.
<https://doi.org/10.1007/s12596-023-01107-y> (Q2, IF=1.6, SCIE)
28. Sachin Singh, A. K. Sharma, Pooja Lohia, D. K. Dwivedi, Vipin Kumar and **Pravin Kumar Singh**, Simulation study of reconfigurable surface plasmon resonance refractive index sensor employing bismuth telluride and MXene nanomaterial for cancer cell detection, *Physica Scripta*, 98 (2023) 025813.
[DOI 10.1088/1402-4896/acb023](https://doi.org/10.1088/1402-4896/acb023) (Q2, IF=2.6, SCIE)
29. **Pravin Kumar Singh**, Ahmad Umar, Sadanand, D. K. Dwivedi, Ahmed A. Ibrahim, Mohsen A. M. Alhamami, and S. Baskoutas, Design and simulation of lead-free perovskite solar cells with a hole transport layer made of NiO nanocomposite, *Science of Advanced Materials* 14 (2022) 1-7.
<https://doi.org/10.1166/sam.2022.4368> (Q3, IF=0.6, Scopus)
30. Shivangani, Pooja Lohia, **Pravin Kumar Singh**, Sachin Singh, & D. K. Dwivedi. Design and modeling of reconfigurable surface plasmon resonance refractive index sensor using Al_2O_3 , nickel, and heterostructure BlueP/WSe₂ nanofilms. *Journal of Optics*, 52 (2023) 1358-1369.
<https://doi.org/10.1007/s12596-022-00973-2> (Q2, IF=1.6, SCIE)
31. Ahmad Umar, Sadanand, **Pravin Kumar Singh**, D. K. Dwivedi, H. Algadi, A. A. Ibrahim, M. A. Alhammai, & S. Baskoutas High Power-Conversion Efficiency of Lead-Free Perovskite Solar Cells: A Theoretical Investigation. *Micromachines*, 13(12), (2022) 2201.
<https://doi.org/10.3390/mi13122201> (Q2, IF=3.0, SCIE)
32. Sachin Singh, A. K. Sharma, Pooja Lohia, D. K. Dwivedi and **Pravin Kumar Singh**, Design and modelling of high-performance surface plasmon resonance refractive index sensor using BaTiO_3 , MXene and nickel hybrid nanostructure. *Plasmonics* 17 (2022) 2049-2062.
<https://doi.org/10.1007/s11468-022-01692-x> (Q2, IF=3.3, SCIE)

33. Vandita Rao, **Pravin Kumar Singh**, Pooja Lohia, D.K. Dwivedi, Non-isothermal crystallization kinetics of $\text{Se}_{82-x}\text{Te}_{18}\text{Ge}_x$ ($0 \leq x \leq 12$) for memory applications, *Indian Journal of Physics*, 96(4) (2022) 1075-1082.
<https://doi.org/10.1007/s12648-021-02036-x> (Q3, IF=1.6, SCIE)
34. Priyanka Jaiswal, Vandita Rao, **Pravin Kumar Singh**, Pooja Lohia, D. K. Dwivedi, How the carbon nanotubes effect the glass transition kinetics and thermal stability of Cu-Se-Te-In chalcogenide glasses, *Journal of Thermal Analysis and Calorimetry*, 147(2) (2022) 1053-1060.
<https://doi.org/10.1007/s10973-020-10485-5> (Q2, IF=3.2, SCIE)
35. Sadanand, **Pravin Kumar Singh**, Shambhavi Rai, Pooja Lohia, D.K. Dwivedi, Comparative study of the CZTS, CuSbS_2 and CuSbSe_2 solar photovoltaic cell with an earth-abundant non-toxic buffer layer, *Solar Energy*, 222 (2021) 175-185.
<https://doi.org/10.1016/j.solener.2021.05.013> (Q1, IF=6.0, SCIE)
36. Shivangi Pandey, Sadanand, **Pravin Kumar Singh**, Pooja Lohia, D.K. Dwivedi, Numerical studies of optimising various buffer layers to enhance the performance of tin sulfide (SnS)-based solar cells, *Transactions on Electrical and Electronic Materials*, 22 (2021) 893-903.
<https://doi.org/10.1007/s42341-021-00311-6> (Q3, IF=1.6, ESCI)
37. Priyanka Jaiswal, **Pravin Kumar Singh**, Pooja Lohia, D. K. Dwivedi, Electrical and optical properties of CNTs additives chalcogenide glasses, *Micro and Nanosystems*, 13 (2021) 146-157.
<https://doi.org/10.2174/1876402912999200623124247> (Q3, IF=1.6, ESCI)
38. Sadanand, Puli Sunny Babu, **Pravin Kumar Singh**, A. K. Thakur, D. K. Dwivedi, Optimization of photovoltaic solar cell performance via the earth abundant Zn_3P_2 back surface field, *Optik*, 229 (2021) 166235.
<https://doi.org/10.2174/1876402912999200623124247> (Q2, IF=3.1, SCIE)
39. Surabhi Mishra, **Pravin Kumar Singh**, Rajesh Yadav, Ahmad Umar, Pooja Lohia, D. K. Dwivedi, Investigation of glass forming ability, linear and non-linear optical properties of Ge-Se-Te-Sb thin films, *Chemical Physics*, 541 (2021) 111021.
<https://doi.org/10.1016/j.chemphys.2020.111021> (Q2, IF=2.0, SCIE)
40. **Pravin Kumar Singh**, D. K. Dwivedi, Fahad A. Alharthi, Adel El Marghany, Optical properties of selenium-rich hybrid thin films containing nano-sized Ge-Se-Te-Sb chalcogenide clusters, *Nanoscience and Nanotechnology Letters* 12 (2020) 1-9.
<https://doi.org/10.1166/nnl.2020.3122> (Q4, IF=1.128, Scopus)
41. Sadanand, **Pravin Kumar Singh**, Pooja Lohia, D. K. Dwivedi, Fahad A. Alharthi, Adel El-Marghany, Nabil Al-Zaqri, Development and Characterization for Solar Photovoltaic Cell Based on $\text{Cu}_2\text{ZnSn}(\text{S}_{0.2}\text{Se}_{0.8})_4$ Thin Film, *Nanoscience and Nanotechnology Letters* 12 (3) (2020) 306-316.
<https://doi.org/10.1166/nnl.2020.3119> (Q4, IF=1.128, Scopus)
42. Sachin Singh, **Pravin Kumar Singh**, Ahmad Umar, Pooja Lohia, Hasan Albargi, L. Castañeda and D. K. Dwivedi, 2D Nanomaterial-Based Surface Plasmon Resonance Sensors for Biosensing Applications, *Micromachines* 11 (2020) 779.
<https://doi.org/10.3390/mi11080779> (Q2, IF=3.0, SCIE)
43. Nitin Rai, Shambhavi Rai, **Pravin Kumar Singh**, Pooja Lohia, D K Dwivedi, Analysis of various ETL materials for an efficient perovskite solar cell by numerical simulation, *Journal of Materials Science: Materials in Electronics*, 31 (2020) 16269-16280.
<https://doi.org/10.1007/s10854-020-04175-z> (Q2, IF=2.8, SCIE)
44. Surabhi Mishra, **Pravin Kumar Singh**, Pooja Lohia and D.K. Dwivedi, Arsenic modified $(\text{Ge}_{11.5}\text{Te}_{12.5}\text{Se}_{67.5})_{100-x}$ compound for IR application, *Journal of Non-Crystalline Solids*, 547 (2020) 120309.
<https://doi.org/10.1016/j.jnoncrysol.2020.120309> (Q2, IF=3.2, SCIE)

45. Priyanka Jaiswal, **Pravin Kumar Singh**, Pooja Lohia, D. K. Dwivedi, Study of ac conductivity mechanism and impedance spectroscopy in CNT added $\text{Cu}_5\text{Se}_7\text{Te}_{10}\text{In}_{10}$ chalcogenide system, *Bulletin of Materials Science*, 43(1) (2020) 1-11.
<https://doi.org/10.1007/s12034-020-02210-7> (Q2, IF=1.9, SCIE)
46. Surabhi Mishra, **Pravin Kumar Singh**, Pooja Lohia and D.K. Dwivedi, A review on doping effect on optical properties of Se-Te glassy alloys and its applications, *Journal of Glass Physics and Chemistry*, 46 (4) (2020) 341-349.
<https://doi.org/10.1134/S1087659620040094> (Q3, IF=1.3, SCIE)
47. Nitesh Shukla, **Pravin Kumar Singh**, H. P. Pathak, and D. K. Dwivedi, Study of structural and optical properties of non-oxide $\text{Se}_{90}\text{Cd}_{10-x}\text{Sb}_x$ ($2 < x < 8$) chalcogenide thin films, *Advanced Science, Engineering and Medicine*, 12 (2020) 116-119.
<https://doi.org/10.1166/asem.2020.2522>
48. **Pravin Kumar Singh**, Nitesh Shukla, Vandita Rao, and D. K. Dwivedi, Effect of Annealing on the Structural and Optical Properties of Amorphous $\text{Ge}_8\text{Se}_{60}\text{Te}_{30}\text{Sb}_2$ Thin Films, *Advanced Science, Engineering and Medicine*, 12 (2020) 31-35.
<https://doi.org/10.1166/asem.2020.2525>
49. Surabhi Mishra, Priyanka Jaiswal, **Pravin Kumar Singh**, Pooja Lohia and D.K. Dwivedi, Effect of CNT doping on optical properties of $\text{Cu}_5\text{Se}_7\text{Te}_{10}\text{In}_{10}$ glassy alloys, *Nanotechnology Perceptions* 15 (2019) 168-173.
<https://doi.org/10.56801/nano-ntp.v15i2.158> (Q4, IF=0.6, Scopus)
50. **Pravin Kumar Singh**, S. K. Sharma, S. K. Tripathi, D. K. Dwivedi, Study of dielectric relaxation and thermally activated a.c. conduction in multicomponent $\text{Ge}_{10-x}\text{Se}_{60}\text{Te}_{30}\text{In}_x$ ($0 \leq x \leq 6$) chalcogenide glasses using CBH model, *Results in Physics*, 12 (2019) 223-236.
<https://doi.org/10.1016/j.rinp.2018.11.048> (Q1, IF=4.4, SCIE)
51. **Pravin Kumar Singh**, Pooja Lohia, D. K. Dwivedi, Investigation of dielectric relaxation and a.c. conductivity of third generation multi-component $\text{Ge}_{10-x}\text{Se}_{60}\text{Te}_{30}\text{Sb}_x$ ($0 \leq x \leq 6$) chalcogenide glasses, *Journal of Materials Science: Materials in Electronics*, 30(14) (2019) 13797-13809.
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66. **Pravin Kumar Singh**, V. Rao, D. K. Dwivedi, 'Recent Trends and Future Prospectus of Smart and Composite Materials' in a 2day (February 24-25, 2017) National Conference on Futuristic Mechanical Engineering Organized by Department of Mechanical Engineering, MMMUT Gorakhpur (U.P.), India.
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Patent

1. Upendra Kulshrestha, **Pravin Kumar Singh** "A Polypropylene Composition and Preparation Method Thereof" Indian Patent Application No.202411079516, filed 19/10/2024 (issued 01/11/2024).
2. Upendra Kulshrestha, **Pravin Kumar Singh**, D. K. Dwivedi "Integrated Perovskite Solar Cell With Triboelectric Nanogenerator For Dual-Mode Energy Harvesting" Indian Patent Application No.202411096938 A, filed 06/12/2024 (issued 27/12/2024).

3. Upendra Kulshrestha, **Pravin Kumar Singh**, D. K. Dwivedi "A Rubber Sheath Material For Diesel Locomotive Cable" Indian Patent Application No. 202411096937 A, filed 06/12/2024 (issued 27/12/2024).

Book/Book Chapter

1. **Pravin Kumar Singh**, Dilip Kumar Dwivedi (Eds.) (2025). *Perovskite Solar Cells: Sustainable Materials, Device Design, Renewable Energy Applications*, Walter de Gruyter GmbH & Co KG, Germany, ISBN:9783111726823. (<https://www.degruyterbrill.com/document/isbn/9783111726847/html>)
2. Katiyar, Pulkit, Singh, Yashwant Kumar, Dwivedi, D.K., Lohia, Pooja and **Singh, Pravin Kumar**. "Chapter 5 Advanced materials for high-efficiency perovskite solar cells". *Perovskite Solar Cells: Sustainable Materials, Device Design, Renewable Energy Applications*, edited by Pravin Kumar Singh and Dilip Kumar Dwivedi, De Gruyter, 2025, pp. 107-138. <https://doi.org/10.1515/9783111726847-005>
3. Pathak, Gauri, Singh, Yashwant Kumar, Dwivedi, D.K. and **Singh, Pravin Kumar**. "Chapter 7 Tandem solar cells: design and performance enhancement". *Perovskite Solar Cells: Sustainable Materials, Device Design, Renewable Energy Applications*, edited by Pravin Kumar Singh and Dilip Kumar Dwivedi, De Gruyter, 2025, pp. 161-178. <https://doi.org/10.1515/9783111726847-007>
4. **Singh, Pravin Kumar**, Pandey, Yash and Kulshrestha, Upendra. "Chapter 12 Importance of perovskite solar cells in sustainable energy solutions". *Perovskite Solar Cells: Sustainable Materials, Device Design, Renewable Energy Applications*, edited by Pravin Kumar Singh and Dilip Kumar Dwivedi, De Gruyter, 2025, pp. 253-286. <https://doi.org/10.1515/9783111726847-012>
5. **Pravin Kumar Singh**, Kirti Panwar, Mukund Kumar (2013). 'Recent Development in Robotics'. ISBN: 978-3-659-39377-8. Lambert Academic Publishing, Germany.
6. Singh, S., **Singh, P.K.**, Sharma, A.K., Lohia, P., Dwivedi, D.K. (2023). Sensitivity Enhancement of Graphene and Blue Phosphorene/MoS₂ Heterostructure-Based SPR Biosensor Using Gold (Au) Metal Layer: Theoretical Insight. In: Mishra, B., Tiwari, M. (eds) VLSI, Microwave and Wireless Technologies. Lecture Notes in Electrical Engineering, vol 877. Springer, Singapore. https://doi.org/10.1007/978-981-19-0312-0_47

International/ National Conferences Organized

1. Worked as convener of "**Powering Progress: International Conference on Energy Functional Materials and Photonics (PPICEFP 2024)**", organized by Department of Physics and Material Science at Madan Mohan Malaviya University of Technology, Gorakhpur held on 1-2 March 2024.
2. Worked as coordinator of **59th Assembly of Advanced Materials Congress** organized by International Association of Advanced Materials, IAAM, Sweden held on 25-27 March 2024 in Stockholm, Sweden
3. Worked as communication secretary of **UN 2023 Water Conference Side Event** organized by International Association of Advanced Materials and People's World Commission on Drought and Flood at Bronx Community College of City University of New York and Columbia University New York, USA on 23 & 24 March 2023 respectively.
4. Worked as a convener of **International Conclave on Materials, Energy and Climate (ICMEC 2023)** jointly organized by IAAM Sweden, IGDTUW New Delhi, IIT Madras, IIT Kharagpur, Rajasthan Vidyapeeth Udaipur, IIW Jodhpur during 18-21 December 2023.
5. Worked as a convener of **International Conclave on Materials, Energy and Climate (ICMEC 2022)** jointly organized by International Association of Advanced Materials, IAAM, Sweden, Indira Gandhi Delhi Technical University for Women, New Delhi, and Shoolini University, Solan during 12-14 December 2022

International/Natio nal Conferences Presentation

1. **Pravin Kumar Singh**, Aanshi Tiwari, Ashutosh Tiwari. "Smart Electrochemical Crystal Nucleation of Active Pharmaceutical Ingredients using Net-Zero Heteropolar Graphene Interface". **Poster presentation**, 65th Assembly of Advanced Materials Congress, in Stockholm, Sweden, (18-20 August 2025).
2. **Pravin Kumar Singh**, Ashutosh Tiwari. "Thermal Characteristics and Prediction of Crystallization Kinetics of Nifedipine Using Calorimetric Study". **Oral presentation**, American Advanced Materials Congress, in Miami, United States of America, (26 February - 01 March 2024).
3. **Pravin Kumar Singh**, Ashutosh Tiwari. "Crystallization Kinetics and Thermal Behaviour of Indomethacin using Differential Scanning Calorimetry". **Oral presentation**, American Advanced Materials Congress, in Orlando, United States of America, (09-12 November 2023).
4. **Pravin Kumar Singh**, Mikael Syväjärvi, Dana Cristina Toncu, Ashutosh Tiwari. "Emerging Energy Harvesting Technology for Portable Self-Powered Systems". **Oral presentation**, American Advanced Materials Congress, organized by International Association of Advanced Materials, IAAM, Sweden (Oct 29-Nov 05, 2022).

5. **Pravin Kumar Singh**, D. K. Dwivedi. "A Correlative Study of Optical Band Gap of Se-Te-M (M =Ag, In, Sb, Hg, Bi) Glassy System for Optical Device Applications". **Oral presentation**, National Conference on "Recent Trends in Mathematical Sciences" NCRTMS-2016, M.M.M. University of Technology, Gorakhpur, India (April 12-13, 2016).
6. **Pravin Kumar Singh**, D. K. Dwivedi. "Effect of annealing on the structural and optical properties of amorphous Ge-Se-Te chalcogenide thin film", **Oral presentation**, International Conference on "Nano Science and Nano Technology" ICNN-2017, Department of Applied Physics, School of Physical Sciences, Babasaheb Bhimrao Ambedkar University, Lucknow, India (Sep. 22-24, 2017)
7. **Pravin Kumar Singh**, D. K. Dwivedi. "Structural analysis of Ge-Se-Te-In nano composite". **Oral presentation**, National Conference on "Soft Matter", Department of Physics, D.D.U. Gorakhpur University Gorakhpur, India (March 27-28, 2018).
8. **Pravin Kumar Singh**, D. K. Dwivedi. "Investigations of structural and optical properties of In doped Ge-Se-Te thin films for semiconducting devices". **Poster presentation**, National symposium on Multidimensional Aspects of Spectroscopy, D.D.U. Gorakhpur University, Gorakhpur, India (Nov. 17-18, 2017).
9. **Pravin Kumar Singh** and D. K. Dwivedi. "Effect of heat treatment on structural and optical properties of $\text{Ge}_{80}\text{Se}_{20}\text{Te}_{30}\text{In}_2$ thin film for optical data storage". **Oral presentation**, XXII Conference of International Academy of Physical Sciences, on 'Emerging Trends in Physical Sciences', R.M.L. Awadh University, Faizabad, India (April 13-15, 2018).
10. **Pravin Kumar Singh**, Surabhi Mishra, Brijesh Kumar and D. K. Dwivedi, "Optical investigations of thermally evaporated $\text{Ge}_{10}\text{Se}_{60}\text{Te}_{30}$ and $\text{Ge}_{80}\text{Se}_{60}\text{Te}_{30}\text{In}_2$ chalcogenide thin film for optical memory devices", **Oral presentation**, 5th IEEE Uttar Pradesh Section International Conference on Electrical, Electronics and Computer Engineering (UPCON), Jointly organised by M.M.M. University of Technology, Gorakhpur (U.P.) India and University of Ryukyus, Okinawa, Japan (Nov 2-4, 2018).
11. **Pravin Kumar Singh**, Vandita Rao, Nitesh Shukla and D. K. Dwivedi, "Physical properties of third generation multicomponent $\text{Ge}_{10-x}\text{Se}_{60}\text{Te}_{30}\text{In}_x$ ($x = 0, 2, 4, 6$) chalcogenide glasses", **Poster presentation**, International Conference on Advanced Materials, Energy and Environmental Sustainability (ICAMEES-2018), organized by Department of Chemistry and Physics, University of Petroleum and Energy Studies, Dehradun, Uttarakhand, India (14-15 Dec 2018).
12. **Pravin Kumar Singh**, Vandita Rao, Pooja Lohia and D. K. Dwivedi, "Study of dielectric relaxation and a.c. conductivity in amorphous $\text{Ge}_{10}\text{Se}_{60}\text{Te}_{30}$ chalcogenide glass", **Oral presentation**, National Symposium on Advanced Materials Science (NSAMS-2018), organized by Department of Physics, DDU Gorakhpur University Gorakhpur, India (December 7- 8, 2018).
13. **Pravin Kumar Singh**, Pooja Lohia and D. K. Dwivedi "Effect of annealing on the structural and optical properties of amorphous $\text{Ge}_{80}\text{Se}_{60}\text{Te}_{20}\text{Sb}_2$ thin films" **Oral presentation**, National Conference on "Smart Materials, Devices and Sustainable Technologies" SMDST-2019, M.M.M. University of Technology, Gorakhpur, India (March 15-16, 2019).
14. **Pravin Kumar Singh**, Pooja Lohia and D. K. Dwivedi, "Investigation of ac conductivity in $\text{Ge}_{80}\text{Se}_{60}\text{Te}_{20}\text{Sb}_2$ quaternary glass", **Oral presentation**, National Conference on "Recent Trends in Devices, Circuits and Communication" RTDC2-2019, M.M.M. University of Technology, Gorakhpur, India (April 15-16, 2019).
15. **Pravin Kumar Singh**, Nitesh Shukla and D. K. Dwivedi, "Effects of thermal annealing on optical properties of Ge-Se- Te-Sb thin films for photonic applications", **Oral presentation**, National E-conference on Interdisciplinary Research in Science and Technology (NCIRST-20) organized by Amiruddaula Islamia Degree College, Lucknow, India (April 30-31, 2020).

Invited Talks

1. Delivered an invited talk on the topic "SMART interface induced crystal nucleation dynamic model for active pharmaceutical ingredients," in the 63rd assembly of the Advanced Materials Congress (AMC) held in Stockholm, Sweden (25-27 March 2025).
2. Delivered an invited talk on the topic "Quantum Assisted SMART APIs: Revolutionizing Drug Discovery, Delivery, and Precision Medicine" at the International Conference on Scope of Quantum Technology in Natural Sciences, Department of Physics & IQAC, Shri Lal Bahadur Shastri Degree College, Gonda, India (28 February-1 March 2025)
3. Delivered an invited talk on the topic "Concept Development and Prototyping" in the AICTE-ATAL Faculty Development Program of Design Thinking For Healthcare Innovation organized by Global Institute of Technology Jaipur, India (17-22 February 2025)
4. Delivered an invited talk on the topic "Triboelectric Nanogenerators for Green Energy Generation" at the One-Day International Symposium on Emerging Trends in Energy and Sustainability (ETES 2024), organized by the Department of Physics, St. Joseph's College (Autonomous), Tiruchirappalli, 620002, India (25 September 2024).

5. Delivered an invited talk on the topic *"Smart Healthcare: Exploring the Next Generation of Medical Devices"* at the International Conference on Translational Research and Innovation (ICTRI 2023), organized by Chengdu University, China (18-19, October 2023).
6. Delivered an invited talk on the topic *"Efficient Healthcare Devices,"* in European Advanced Materials Congress (EAMC), Southampton, UK (23-26 September, 2023).
7. Delivered an invited talk on the topic *"Nanogenerators for Green Energy Scavenging Applications"* at the International Conference on Translational Research and Innovation, a special focus on Green Energy, Materials, and Healthy Sustainable Developments, organized by Chengdu University, China (7-8 September 2022).
8. Delivered an invited talk on the topic *"Prognosticating Crystallization Kinetics in Active Pharmaceutical Ingredients"* at the European Assembly Advanced Materials Congress held in Stockholm, Sweden (28-31 August 2022).
9. Delivered an invited talk on the topic *"Nanostructured Piezoelectric Energy Harvesters,"* in the 8th assembly of the European Advanced Materials Congress (EAMC) held at MSC Seaside Cruise, Italy (June 25- July 02, 2022).
10. Delivered a lecture in the webinar on 7th Anniversary and Foundation Day on the topic *"Recent Progress and Development Trends in Material Science"* organized by Department of Physics and Material Science, MMM University of Technology Gorakhpur (Dec 1, December 2020).

Reviewer and Editorial Activities

- **Reviewer for:** Solar Energy Materials and Solar Cells, Solar Energy, Optical Materials, Journal of Alloys and Compound, Journal of Physics and Chemistry of Solids, Optical Fibre Technology, Material Science and Engineering B, Journal of Optics etc.
- **Managing Editor** for Advanced Materials Proceeding, International Association of Advanced Materials, Europe, **Associate Editor** for ScieEng Advances, Ariston Publications and **Guest Editor** for Sensor Letters 17, 784–786 (2019).

Research Guidance

International Pre-PhD Students

(Cosupervised following students during group leader tenure)

- Research assignment entitled, *"John B. Goodenough's pioneering contributions towards advancements in photo-rechargeable lithium batteries"*, Soumyoraj Mallick (Pre. PhD completed in 2024).
<https://www.sciencedirect.com/science/article/abs/pii/S2211285524005408>
- Research assignment entitled, *"Wurtzite nanostructured piezoelectric devices"*, Xuli Cheng, (Pre. PhD completed in 2023). <https://www.sciencedirect.com/science/article/abs/pii/S2589234723001604>
- Research assignment entitled *"Trends in smart drug delivery systems for targeting cancer cells"*, Sharanya Kandula (Pre. PhD completed in 2023).
<https://www.sciencedirect.com/science/article/abs/pii/S0921510723005585>
- Research assignment entitled *"Bio-capture of greenhouse gases as a reliable and cost-effective method to achieve carbon neutrality,"* Tanmay Tiwari (Pre. PhD completed in 2023).
<https://www.sciencedirect.com/science/article/abs/pii/S0048969724025798>
- Research assignment entitled *"Simulation and modelling of inorganic CsGeI₃ on Si-tandem solar cell"* Sujata, (Pre. PhD completed in 2023).

PhD and Master Students

(Guided following PhD and master students with instruction of my supervisor during PhD and Guest Faculty tenure)

- Thesis entitled *"Development and electrical characterization of CNT additive Se-based chalcogenide glasses"*, Priyanka Jaiswal (PhD awarded on 10.02.2020)
- Thesis entitled *"Development and optical characterization of Se-based chalcogenide glasses for sensor application"*, Surbhi Mishra (PhD awarded on 08.04.2021)
- Thesis entitled *"Development & characterization of CZTS based solar cell"* by Sadanand (PhD awarded on 26.10.2021)
- Thesis entitled *"Thin Film Characterization and Its Application in Electronic Devices and Circuits"*, Sonam Tripathi (PhD awarded on 01.02.2022)
- Thesis entitled *"Development & characterization of bulk heterojunction organic solar cell"*, Shambhavi Rai (PhD awarded on 01.02.2022)
- Project dissertation entitled *"Modelling of four-layer planer waveguide grating for sensing"*, Adarsh Chandra Mishra (M.Sc. completed in 2020)
- Project dissertation entitled *"Effect of tailored ETL response on perovskite solar cell structure"*, Nitin Rai (M.Sc. completed in 2020)

- ## Membership

- ## Research Highlights

Reference

1. Dr. Anshuman Mishra, Group Leader, Institute of Advanced Materials, IAAM, Sweden (E-mail: anshuman.mishra@iaam.se) Mob: +46-739299485
2. Prof. D. K. Dwivedi Dean of Academic Affairs, MMM University of Technology Gorakhpur, UP, India. (E-mail: dkdas@mmmut.ac.in) Mob: +91-9235500510
3. Prof. Sanjay Tiwari, Vice Chancellor, Madhya Pradesh Bhoj Open University, M.P. (E-mail: drsanjaytiwari@gmail.com). Mob: +91-9424225771
4. Mr. M. P. Kamath, Head, Optical Design and Development Lab Nd:Glass Laser Laboratory, RRCAT, Indore, India (E-Mail: kamath@rrcat.gov.in) Ph.No: +91-7312442633

Other Information

Community service

Place: Ulrika, Sweden

Gerwin

(Dr. Pravin Kumar Singh)